

se assumo $q_2 = 0 \Leftrightarrow$ solo primo modo

$$\beta^{(i)}(q_1, q_2) = c_0^{(i)} + c_1^{(i)} q_1 + c_2^{(i)} q_2 + c_3^{(i)} q_1^2 + c_4^{(i)} q_1 q_2 + c_5^{(i)} q_2^2 + c_6^{(i)} q_1^3 + c_7^{(i)} q_1^2 q_2 + c_8^{(i)} q_1 q_2^2 + c_9^{(i)} q_2^3, \quad (5)$$

$$\beta^{(1)}(\bar{q}) = c_0^{(1)} + c_1^{(1)} q_1 + c_3^{(1)} q_1^2 + c_6^{(1)} q_1^3 \approx c_0^{(1)} + c_1^{(1)} q_1 \quad \text{per } q_1 \text{ piccoli}$$

$$\ddot{q}_1 + \frac{\omega_{01}}{Q_1} \dot{q}_1 + \beta^{(1)}(q_1) V_{DC}^2 - 2 V_{DC} V_{AC} F_{DA}^{(1)} \sin(\omega t) = 0$$

$$s U_1 + \frac{\omega_{01}}{Q_1} U_1 + c_0^{(1)} + c_1^{(1)} \frac{U_1}{s} = f_{DB}^{(1)} V_{DC}^2 + 2 V_{DC} V_{in} \cdot f_{DA}^{(1)}$$

$$F_{DB}^{(1)}(\bar{q}) \approx c_{0,DB}^{(1)} + c_{1,DB}^{(1)} q_1$$

$$f_{DA}^{(1)} \approx c_{0,DA}^{(1)} + c_{1,DA}^{(1)} q_1$$

$$\hookrightarrow \ddot{q}_1 + \frac{\omega_{01}}{Q_1} \dot{q}_1 + c_0^{(1)} + c_1^{(1)} q_1 = c_{0,DB}^{(1)} V_{DC}^2 + c_{1,DB}^{(1)} q_1 V_{DC}^2 + (c_{0,DA}^{(1)} + c_{1,DA}^{(1)} q_1) 2 V_{DC} V_{in}$$

$$\hookrightarrow \ddot{q}_1 + \frac{\omega_{01}}{Q_1} \dot{q}_1 + \left(c_0^{(1)} - c_{1,DA}^{(1)} \cdot 2 V_{DC} V_{in} \right) q_1 + F_c = c_{0,DA}^{(1)} \cdot 2 V_{DC} V_{in}$$

$$\hookrightarrow s U_1 + \frac{\omega_{01}}{Q_1} U_1 + c_1^{(1)} \frac{U_1}{s} + F_c = 2 c_{1,DA}^{(1)} \mathcal{L}(q_1 \cdot V_{in}) + c_{0,DA}^{(1)} \cdot 2 V_{DC} V_{in}$$

se V_{in} e q_1 sono due cos :

magari si può dire che è piccolo $\Rightarrow$ trascurabile, anche perché mi darebbe componenti a f doppia e continue.

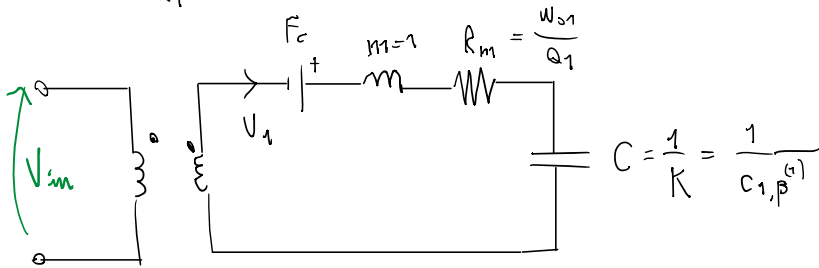
$$\frac{V_{AC} \cdot Q_{max}}{2} \left(1 + \cos(2\omega_{01} t) \right) \quad \text{se in fase}$$

$$\frac{V_{AC} Q_{max}}{2} \left(\cos(\Delta\varphi) + \cos(2\omega_{01} t + \Delta\varphi) \right)$$

$$\frac{V_{AC} Q_{max}}{2} \left(\cos(\Delta\varphi) + \cos(2\omega_{01}t + \Delta\varphi) \right)$$

Non tutte le approx:

$$sV_1 + \frac{\omega_{01}}{Q_1} V_1 + c_{1,p}^{(1)} \frac{V_1}{s} = c_{0,DA}^{(1)} \cdot 2V_{DC} \cdot V_{in} + F_c$$



$$\alpha = c_{0,DA}^{(1)} \cdot 2V_{DC}$$

$$F_c = c_{0,DD}^{(1)} \cdot V_{DC}^2 - c_{0,p}^{(1)} \quad \text{si può annullare?}$$

$$= 96.08 \cdot \epsilon_0 \cdot V_{DC}^2 + 9.72 \cdot 10^{-9} \quad \text{NO, Fuck}$$

Table 2 Coefficients of the electrostatic nonlinear manifold numerically computed for the two modes under study

	$f_{DD}^{(1)}/\epsilon_0$	$f_{DA}^{(1)}/\epsilon_0$ (i = 1)	$f_{DT}^{(1)}/\epsilon_0$	$f_{TT}^{(1)}/\epsilon_0$	$f_{DD}^{(2)}/\epsilon_0$	$f_{DA}^{(2)}/\epsilon_0$ (i = 2)	$f_{DT}^{(2)}/\epsilon_0$	$f_{TT}^{(2)}/\epsilon_0$
$c_0^{(i)}$	96.08	-91.81	-94.50	93.94	-41.01	-202.73	34.82	-34.79
$c_1^{(i)}$	21.11	-5.52	-10.27	10.27	8.34	-6.05	3.88	-3.89
$c_2^{(i)}$	8.51	-6.10	3.80	-3.80	37.76	-17.62	-1.63	1.63
$c_3^{(i)}$	1.28	-0.38	-1.26	1.26	-0.51	-0.32	0.50	-0.50
$c_4^{(i)}$	-1.06	-0.60	1.05	-1.05	0.41	-1.05	-0.46	0.46
$c_5^{(i)}$	0.25	-0.51	-0.28	0.28	-0.25	-1.62	0.13	-0.13
$c_6^{(i)}$	0.14	-0.02	-0.11	0.11	-0.02	-0.01	0.04	-0.04
$c_7^{(i)}$	-0.06	-0.04	0.13	-0.13	0.16	-0.05	-0.06	0.06
$c_8^{(i)}$	0.16	-0.05	-0.06	0.06	0.19	-0.11	0.03	-0.03
$c_9^{(i)}$	0.059	-0.03	0.01	-0.01	0.25	-0.12	-0.01	0.01

Numerical values of the $c_j^{(i)}$ coefficients are consistent with q_k expressed in [μm] and $f_{ab}^{(i)}/\epsilon_0$ in [$\mu\text{N}\mu\text{m}/(\text{V}^2 \text{pF})$]

	$\beta^{(1)} [\mu\text{m}/\mu\text{s}^2]$ (i = 1)	$\beta^{(2)} [\mu\text{m}/\mu\text{s}^2]$ (i = 2)
$c_0^{(i)}$	$-9.72 \cdot 10^{-9}$	$-1.34 \cdot 10^{-7}$
$c_1^{(i)}$	6.85	$-2.23 \cdot 10^{-6}$
$c_2^{(i)}$	$1.08 \cdot 10^{-6}$	27.47
$c_3^{(i)}$	$9.55 \cdot 10^{-2}$	$-5.52 \cdot 10^{-2}$
$c_4^{(i)}$	-0.11	0.20
$c_5^{(i)}$	$9.98 \cdot 10^{-2}$	0.40
$c_6^{(i)}$	$3.62 \cdot 10^{-4}$	$-4.86 \cdot 10^{-4}$
$c_7^{(i)}$	$-1.46 \cdot 10^{-3}$	$3.00 \cdot 10^{-3}$
$c_8^{(i)}$	$3.00 \cdot 10^{-3}$	$-7.51 \cdot 10^{-3}$
$c_9^{(i)}$	$-2.52 \cdot 10^{-3}$	$1.01 \cdot 10^{-2}$

Numerical values of the $c_j^{(i)}$ coefficients are consistent with q_k expressed in [μm] and $\beta^{(i)}$ in [$\mu\text{m}/\mu\text{s}^2$]

$$\frac{V_1}{V_{in}} = \frac{c_{0,DA}^{(1)} \cdot 2V_{DC}}{s + \frac{\omega_{01}}{Q_1} + \frac{c_{0,p}^{(1)}}{s}} = \frac{s \cdot c_{0,DA}^{(1)} \cdot 2V_{DC}}{s^2 + \frac{\omega_{01}}{Q_1} s + c_{0,p}^{(1)}}$$

$\omega_r \neq \omega_{01}$! ✓

magari per questo
che il max di ampiezza
di I_s non è a ω_{01}

$$I(\mathbf{q}) = \sum_{i=1}^2 \left(\int_{S_1} \frac{\tilde{\sigma}_4 \tilde{\sigma}_1}{2\epsilon_0} \boldsymbol{\psi}_i \cdot \mathbf{n} dS \right) V_{DC} \dot{q}_i$$

$$\simeq \sum_{i=1}^2 f_{DA}^{(i)}(\mathbf{q}) V_{DC} \dot{q}_i$$

sempre ipotesi solo primo modo
 $\tilde{q}(\bar{\varphi}) \simeq f_{DA}^{(1)}(\bar{\varphi}) V_{DC} \dot{q}_1$

$$\sum_{i=1}^2 \nabla J_{S1} \angle \epsilon_0$$

$$\simeq \sum_{i=1}^2 f_{DA}^{(i)}(\mathbf{q}) V_{DC} \dot{q}_i$$

$$\begin{aligned} i(\bar{q}) &\simeq f_{DA}^{(1)}(\bar{q}) V_{DC} \dot{q}_1 \\ &\simeq C_{0,DA}^{(1)} V_{DC} \dot{q}_1 + C_{1,DA}^{(1)} V_{DC} q_1 \cdot \dot{q}_1 \end{aligned}$$

$\hookrightarrow I \simeq C_{0,DA}^{(1)} \cdot V_{DC} U_1 + C_{1,DA}^{(1)} V_{DC} \mathcal{L}(q_1 \cdot \dot{q}_1)$
 ho una componente a ω_0 e una a $2\omega_0$
SE ho solo ω_0 in q_1

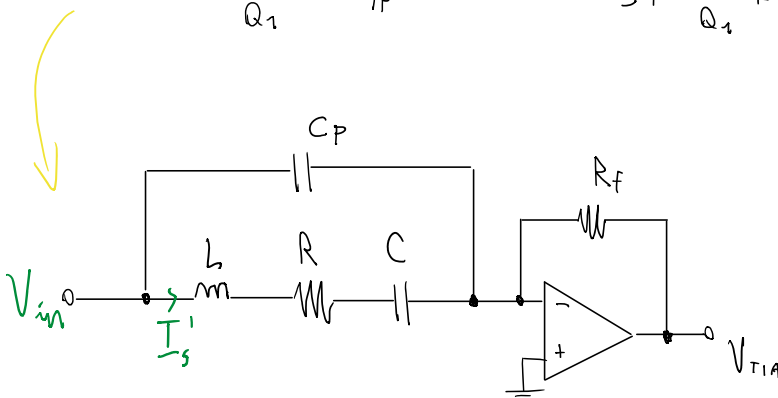
se $q_1 = Q_{max} \cos(\omega_0 t)$ $\leadsto q_1 \cdot \dot{q}_1 = Q_{max}^2 \cdot \omega_0 \cos(\omega_0 t) (-\sin(\omega_0 t))$

$$\begin{aligned} &= Q_{max}^2 \cdot \omega_0 \cos(\omega_0 t) \cos(\omega_0 t + \frac{\pi}{2}) \\ &= Q_{max}^2 \cdot \omega_0 \cdot \frac{1}{2} \left(\cos\left(\frac{\pi}{2}\right) + \cos(2\omega_0 t + \frac{\pi}{2}) \right) \\ &= - \frac{Q_{max}^2 \cdot \omega_0}{2} \sin(2\omega_0 t) \end{aligned}$$

comunque $Q^2 \cdot \omega_0 \sim (10^{-9})^2 \cdot 10^6 \sim 10^{-12}$
 trascurabile

Componente lineare con V_{in} :

$$\begin{aligned} I_s' &= C_{0,DA}^{(1)} \cdot V_{DC} U_1 = C_{0,DA}^{(1)} \cdot V_{DC} \cdot \frac{s \cdot C_{0,DA}^{(1)} \cdot 2 V_{DC}}{s^2 + \frac{\omega_{01}}{Q_1} s + C_{0,\beta}^{(1)}} V_{in} \\ &= \frac{s (C_{0,DA}^{(1)} V_{DC})^2 \cdot 2}{s^2 + \frac{\omega_{01}}{Q_1} s + C_{0,\beta}^{(1)}} V_{in} = \frac{2 (C_{0,DA}^{(1)} \cdot V_{DC})}{s + \frac{\omega_{01}}{Q_1} + \frac{C_{0,\beta}^{(1)}}{s}} \end{aligned}$$



$$L = \frac{1}{2\alpha^2} \quad R = \frac{\omega_0}{Q_1 \cdot 2\alpha^2} \quad C = 2\alpha^2 \cdot C_{0,\beta}^{(1)} \quad \mathcal{L} = C_{0,DA}^{(1)} V_{DC}$$

Da qui si potrebbe fare lo stesso ragionamento e esprimere C_p dello riferimento

Cause possibili la distorsione (non ha sinusoide pura):

- termini non lineari introdotti da $\beta, f_{DA}^{(i)}, f_{DD}^{(i)}$
- termine $q_i \cdot \dot{q}_i$ nella formula della corrente e $V_{in} \cdot q$ nel modello
- eccitazione del secondo modo ω_2 che introduce anche i termini in q_2